

Decision Trees, Random Forests

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Decision Tree basics

DT from predictor space, and *vice versa*

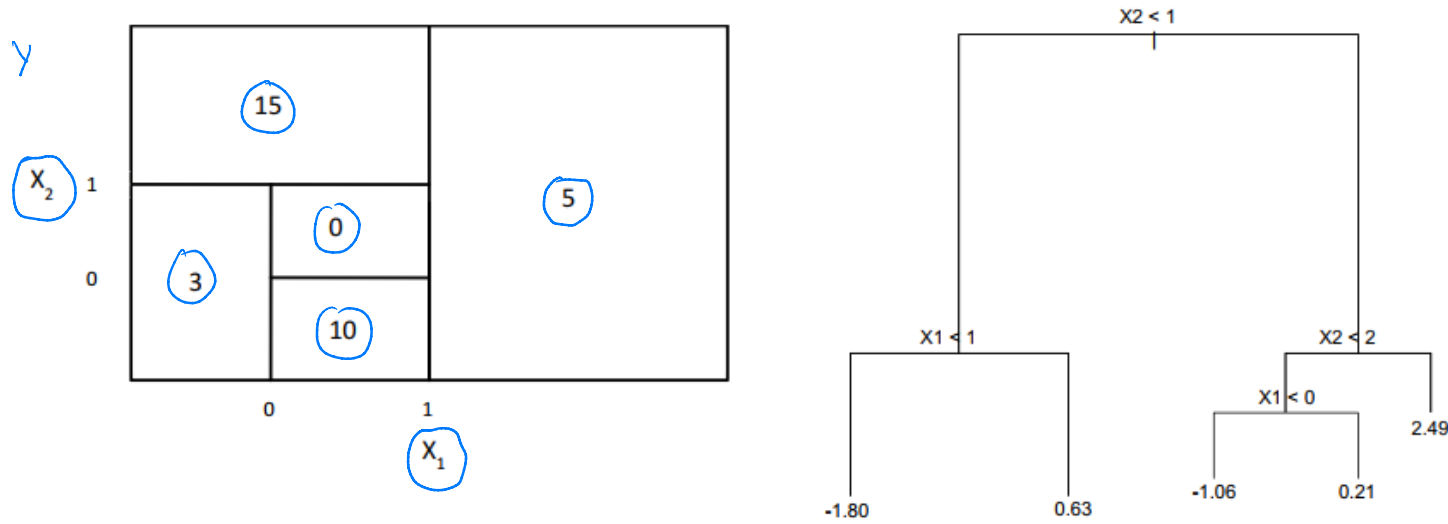
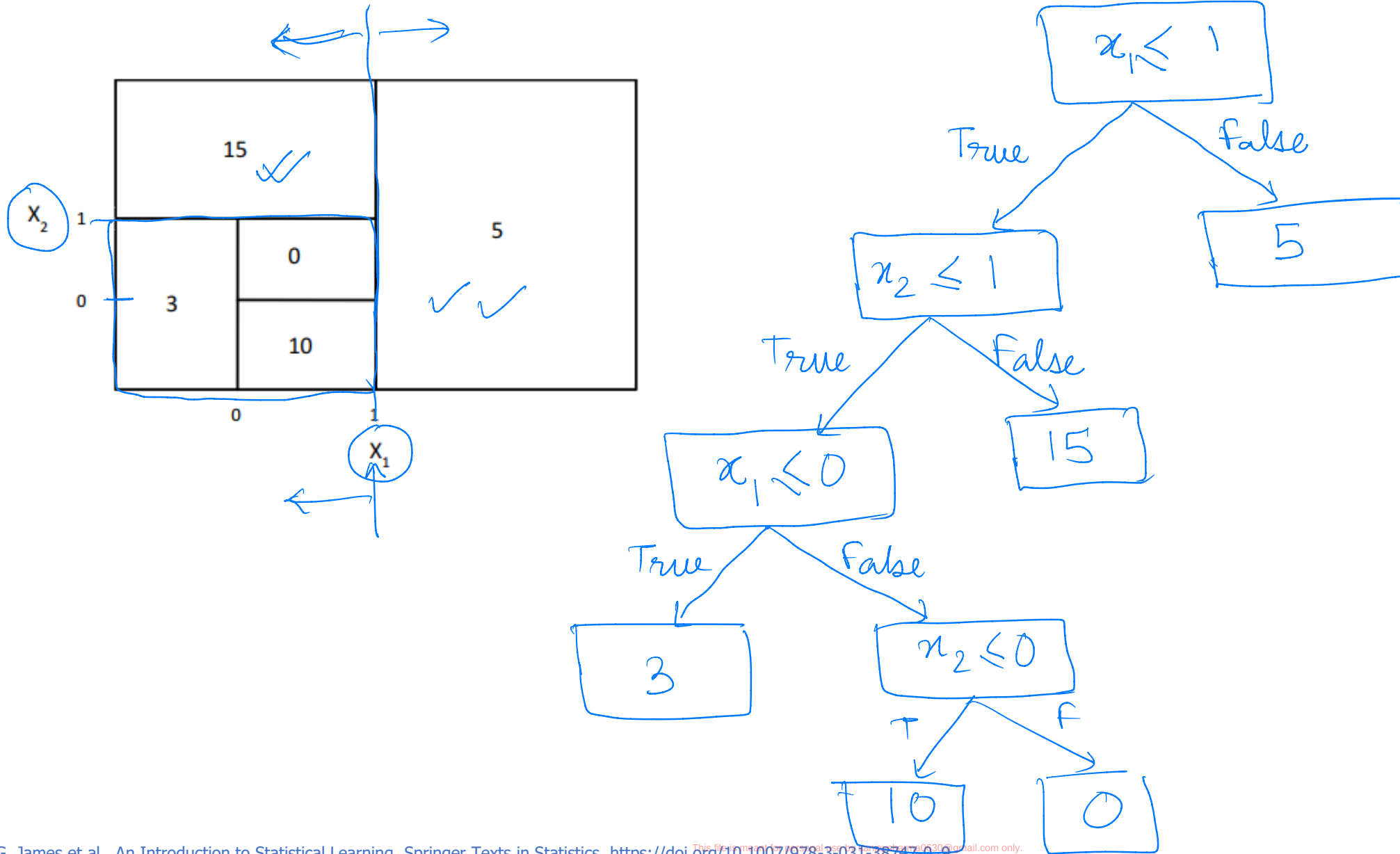


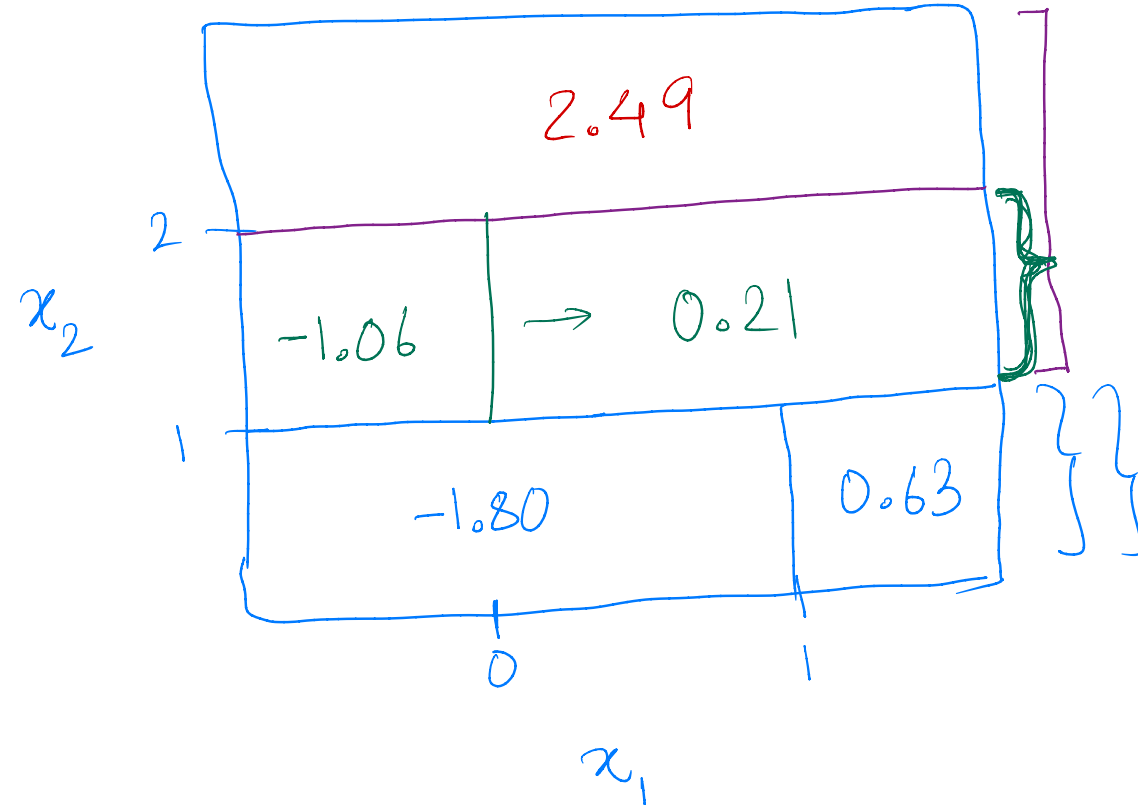
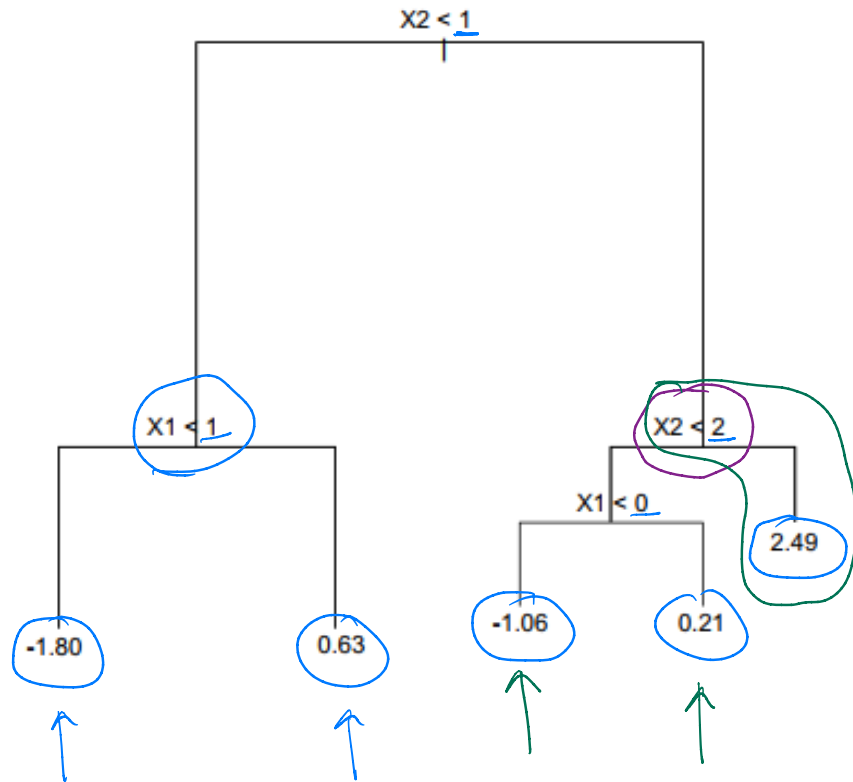
FIGURE 8.14. Left: A partition of the predictor space corresponding to Exercise 4a. Right: A tree corresponding to Exercise 4b.

- Sketch the tree corresponding to the partition of the predictor space illustrated in the left-hand panel of Figure 8.14. The numbers inside the boxes indicate the mean of Y within each region.
- Create a diagram similar to the left-hand panel of Figure 8.14, using the tree illustrated in the right-hand panel of the same figure. You should divide up the predictor space into the correct regions, and indicate the mean for each region.

DT from predictor space, and *vice versa*



DT from predictor space, and *vice versa*



Gini Impurity, Root node

Gini impurity, Choice of root node



Consider building a decision tree to classify the following dataset. The features are Outlook $\in \{\text{Sunny, Overcast}\}$ and Humidity $\in \{\text{High, Normal}\}$, and the label is Play $\in \{\text{Yes, No}\}$:

Outlook	Humidity	Play
Sunny	High	No
Sunny	Normal	Yes
Overcast	High	Yes
Overcast	Normal	Yes
Sunny	High	No

- (a) Compute the Gini impurity, $G(\text{Play})$, of the full dataset
- (b) Determine and justify the choice of feature for the root node.

Gini impurity, Choice of root node



Outlook	Humidity	Play
Sunny	High	No
Sunny	Normal	Yes
Overcast	High	Yes
Overcast	Normal	Yes
Sunny	High	No

No. of samples, $N = 5$

3 Yes , 2 No

$$p(\text{Yes}) = \frac{3}{5}$$

$$p(\text{No}) = \frac{2}{5}$$

(a) Compute the Gini impurity, $G(\text{Play})$, of the full dataset

The Gini impurity for a node S with c classes is calculated as:

$$G(S) = 1 - \sum_{i=1}^c p_i^2$$

$$G(S) = 1 - \sum_{i=1}^c p_i^2$$

$$= 1 - \left\{ \left(\frac{3}{5}\right)^2 + \left(\frac{2}{5}\right)^2 \right\}$$

$$= 1 - \frac{13}{25} = \frac{12}{25} = 0.48$$

Gini impurity, Choice of root node



Outlook	Humidity	Play
Sunny	High	No
Sunny	Normal	Yes
Overcast	High	Yes
Overcast	Normal	Yes
Sunny	High	No

(a) Compute the Gini impurity, $G(Play)$, of the full dataset

The Gini impurity for a node S with c classes is calculated as:

$$G(S) = 1 - \sum_{i=1}^c p_i^2$$

Gini impurity, Choice of root node



Outlook	Humidity	Play
Sunny	High	No
Sunny	Normal	Yes
Overcast	High	Yes
Overcast	Normal	Yes
Sunny	High	No

(b) Determine and justify the choice of feature for the root node.

Weighted Average Gini Impurity for a Split on Feature (A):

$$G(S, A) = \sum_{j=1}^v \frac{|S_j|}{|S|} G(S_j)$$

Where v is the number of distinct values of feature A , S_j is the subset of samples for value j , and $|S|$ is the total number of samples in the parent node. **Our goal is to minimize this value at each split.**

Feature Outlook — Subset Sunny, Subset Overcast, $N=5$

Sunny (samples = 3)

$$p(\text{Yes}) = \frac{1}{3}$$

$$p(\text{No}) = \frac{2}{3}$$

$$G(\text{Sunny}) = 1 - \sum_{i=1}^c p_i^2 = 1 - \left\{ \left(\frac{1}{3} \right)^2 + \left(\frac{2}{3} \right)^2 \right\} = 1 - \frac{5}{9} = \frac{4}{9}$$

Gini impurity, Choice of root node



Outlook	Humidity	Play
Sunny	High	No
Sunny	Normal	Yes
Overcast	High	Yes ✓
Overcast	Normal	Yes ✓
Sunny	High	No

(b) Determine and justify the choice of feature for the root node.

Weighted Average Gini Impurity for a Split on Feature (A):

$$G(S, A) = \sum_{j=1}^v \frac{|S_j|}{|S|} G(S_j)$$

Where v is the number of distinct values of feature A , S_j is the subset of samples for value j , and $|S|$ is the total number of samples in the parent node. **Our goal is to minimize this value at each split.**

Subset Overcast

samples = 2

$p(\text{Yes}) = \frac{2}{2}$

$p(\text{No}) = \frac{0}{2}$

$$G(\text{Overcast}) = 1 - \left\{ \left(\frac{2}{2}\right)^2 + \left(\frac{0}{2}\right)^2 \right\} = 1 - 1 = 0$$

Gini impurity, Choice of root node



Outlook	Humidity	Play
Sunny	High	No
Sunny	Normal	Yes
→ Overcast	High	Yes
→ Overcast	Normal	Yes
Sunny	High	No

(b) Determine and justify the choice of feature for the root node.

Weighted Average Gini Impurity for a Split on Feature (A):

$$G(S, A) = \sum_{j=1}^v \frac{|S_j|}{|S|} G(S_j)$$

Where v is the number of distinct values of feature A , S_j is the subset of samples for value j , and $|S|$ is the total number of samples in the parent node. **Our goal is to minimize this value at each split.**

$$\begin{aligned} G(S, \text{Outlook}) &= \frac{3}{5} \times G(\text{Sunny}) + \frac{2}{5} \times G(\text{Overcast}) \\ &= \frac{3}{5} \times \frac{4}{9} + \frac{2}{5} \times 0 = \frac{4}{15} = 0.2667 \end{aligned}$$

Gini impurity, Choice of root node



Outlook	Humidity	Play
Sunny	High ✓	No
Sunny	Normal	Yes
Overcast	High ✓	Yes
Overcast	Normal	Yes
Sunny	High ✓	No

(b) Determine and justify the choice of feature for the root node.

Weighted Average Gini Impurity for a Split on Feature (A):

$$G(S, A) = \sum_{j=1}^v \frac{|S_j|}{|S|} G(S_j)$$

Where v is the number of distinct values of feature A , S_j is the subset of samples for value j , and $|S|$ is the total number of samples in the parent node. **Our goal is to minimize this value at each split.**

Feature Humidity — Subset High, Subset Normal, ($N=5$)

Subset High

Samples = 3

$$p(\text{Yes}) = \frac{1}{3}, \quad p(\text{No}) = \frac{2}{3}$$

$$\begin{aligned} G(\text{High}) &= 1 - \sum_{i=1}^C p_i^2 = 1 - \left\{ \left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 \right\} \\ &= 1 - \frac{5}{9} = \frac{4}{9} \end{aligned}$$

Gini impurity, Choice of root node



Outlook	Humidity	Play
Sunny	High	No
Sunny	Normal	Yes
Overcast	High	Yes
Overcast	Normal	Yes
Sunny	High	No

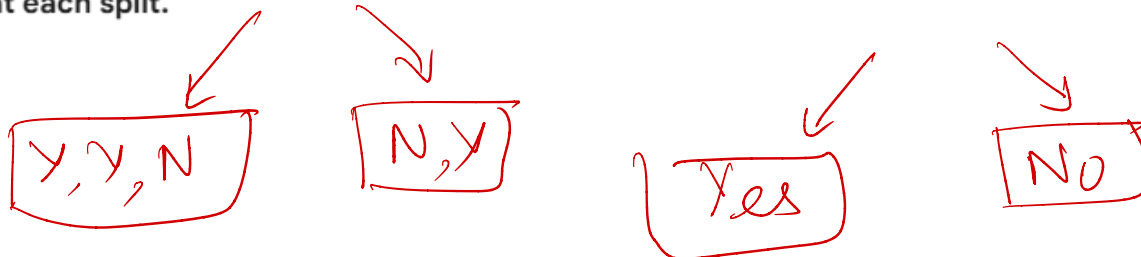
(b) Determine and justify the choice of feature for the root node.

Weighted Average Gini Impurity for a Split on Feature (A):

$$G(S, A) = \sum_{j=1}^v \frac{|S_j|}{|S|} G(S_j) \quad \downarrow \text{Lower is good}$$

Where v is the number of distinct values of feature A , S_j is the subset of samples for value j , and $|S|$ is the total number of samples in the parent node. **Our goal is to minimize this value at each split.**

$$\frac{\text{Subset Normal}}{G(\text{Normal})} = 0$$



$$G(S, \text{Humidity}) = \frac{3}{5} \times \frac{4}{9} + \frac{2}{5} \times 0 = \frac{4}{15} = 0.2667$$

$$G(S, \text{Outlook}) = 0.2667$$

Gini impurity, Choice of root node



Outlook	Humidity	Play
Sunny	High	No
Sunny	Normal	Yes
Overcast	High	Yes
Overcast	Normal	Yes
Sunny	High	No

(b) Determine and justify the choice of feature for the root node.

Weighted Average Gini Impurity for a Split on Feature (A):

$$G(S, A) = \sum_{j=1}^v \frac{|S_j|}{|S|} G(S_j)$$

Where v is the number of distinct values of feature A , S_j is the subset of samples for value j , and $|S|$ is the total number of samples in the parent node. **Our goal is to minimize this value at each split.**

Gini impurity, Choice of root node



Outlook	Humidity	Play
Sunny	High	No
Sunny	Normal	Yes
Overcast	High	Yes
Overcast	Normal	Yes
Sunny	High	No

(b) Determine and justify the choice of feature for the root node.

Weighted Average Gini Impurity for a Split on Feature (A):

$$G(S, A) = \sum_{j=1}^v \frac{|S_j|}{|S|} G(S_j)$$

Where v is the number of distinct values of feature A , S_j is the subset of samples for value j , and $|S|$ is the total number of samples in the parent node. **Our goal is to minimize this value at each split.**

Decision Tree example

Decision Tree, Feature selection



Construct the decision tree to help determine whether a child should go out to play.

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No

6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



No. of samples, $N = 10$

Features $\rightarrow 4$

Weather $\rightarrow$ Sunny₃, Cloudy₃, Rainy₄

Temperature $\rightarrow$ Hot₄, Mild₅, Cool₁

Humidity $\rightarrow$ High₇, Normal₃

Wind $\rightarrow$ Weak₄, Strong₆

5 Yes, 5 No

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 0: Calculate base Gini impurity

$$G(s) = 1 - \sum_{i=1}^c p_i^2$$

$$p(\text{Yes}) = \frac{5}{10}, \quad p(\text{No}) = \frac{5}{10}$$

$$\begin{aligned} G(s) &= 1 - \left\{ \left(\frac{5}{10}\right)^2 + \left(\frac{5}{10}\right)^2 \right\} \\ &= 1 - \left\{ (0.5)^2 + (0.5)^2 \right\} \\ &= 0.50 \end{aligned}$$

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 1: Determine the root split – Weather/Temperature/Humidity/Wind?

Feature: Weather

Subset : Sunny (3)

$$P(\text{Yes}) = \frac{1}{3}, P(\text{No}) = \frac{2}{3}$$

$$G(\text{Sunny}) = 1 - \left\{ \left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 \right\} = 1 - \frac{5}{9} = \frac{4}{9}$$

Subset : Cloudy (3)

$$P(\text{Yes}) = \frac{3}{3}, P(\text{No}) = \frac{0}{3}$$

$$G(\text{Cloudy}) = 0$$

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 1: Determine the root split – Weather/Temperature/Humidity/Wind?

Feature: Weather

Subset Rainy (4)

$$P(\text{Yes}) = \frac{1}{4}, \quad P(\text{No}) = \frac{3}{4}$$

$$G(\text{Rainy}) = 1 - \left\{ \left(\frac{1}{4}\right)^2 + \left(\frac{3}{4}\right)^2 \right\}$$
$$= 1 - \frac{10}{16} = \frac{3}{8}$$

$$G(S, \text{Weather}) = \frac{3}{10} \times G(\text{Sunny}) + \frac{3}{10} \times G(\text{Cloudy})$$
$$+ \frac{4}{10} \times G(\text{Rainy})$$
$$= \frac{3}{10} \times \frac{4}{9} + \frac{3}{10} \times 0 + \frac{4}{10} \times \frac{3}{8}$$
$$= 0.1333 + 0 + 0.15 = 0.2833$$

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 1: Determine the root split – Weather/Temperature/Humidity/Wind?

Feature: Weather

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 1: Determine the root split – Weather/Temperature/Humidity/Wind?

Feature: Temperature

Subset Hot (4)

$$P(\text{Yes}) = \frac{2}{4}, P(\text{No}) = \frac{2}{4}$$

$$G(\text{Hot}) = 1 - \left\{ \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 \right\} = 0.50$$

Subset Mild (5)

$$P(\text{Yes}) = \frac{3}{5}, P(\text{No}) = \frac{2}{5}$$

$$\begin{aligned} G(\text{Mild}) &= 1 - \left\{ \left(\frac{3}{5}\right)^2 + \left(\frac{2}{5}\right)^2 \right\} \\ &= 1 - \frac{13}{25} = \frac{12}{25} \end{aligned}$$

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 1: Determine the root split – Weather/Temperature/Humidity/Wind?

Feature: Temperature

Subset Cool (1)

$$P(\text{Yes}) = 0, P(\text{No}) = 1$$

$$G(\text{Cool}) = 0$$

$$G(S, \text{Temperature}) = \frac{4}{10} \times G(\text{Hot}) + \frac{5}{10} \times G(\text{Mild}) + \frac{1}{10} \times G(\text{Cool})$$

$$= 0.4 \times 0.5 + 0.5 \times \frac{12}{25} + 0$$

$$= 0.2 + 0.24 = 0.44$$

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 1: Determine the root split – Weather/Temperature/Humidity/Wind?

Feature: Humidity

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 1: Determine the root split – Weather/Temperature/Humidity/Wind?

Feature: Humidity

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 1: Determine the root split – Weather/Temperature/Humidity/Wind?

Feature: Wind

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 1: Determine the root split – Weather/Temperature/Humidity/Wind?

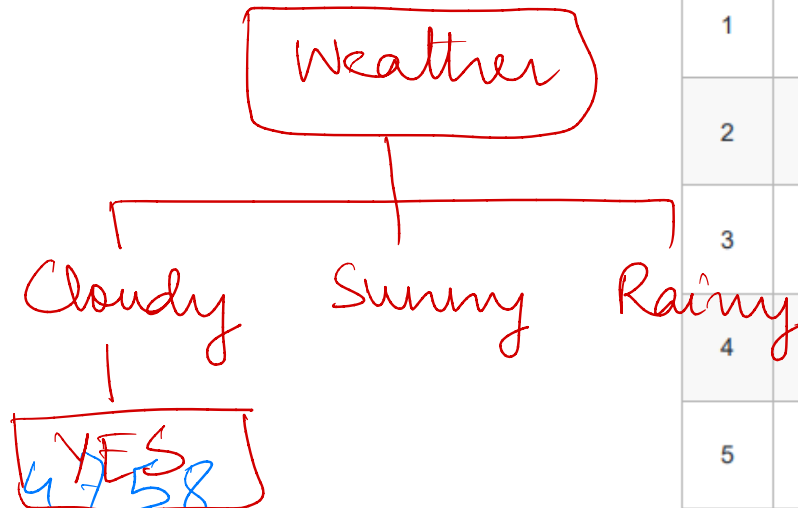
Feature: Wind

$$G(S, \text{Weather}) = 0.2833$$

$$G(S, \text{Temp}) = 0.44$$

$$G(S, \text{Humidity}) = 0.4758$$

$$G(S, \text{Wind}) = 0.417$$



Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 2: Internal nodes – Weather = Cloudy

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 2: Internal nodes – Weather = Sunny

Feature Temperature

Subset : Hot (2)

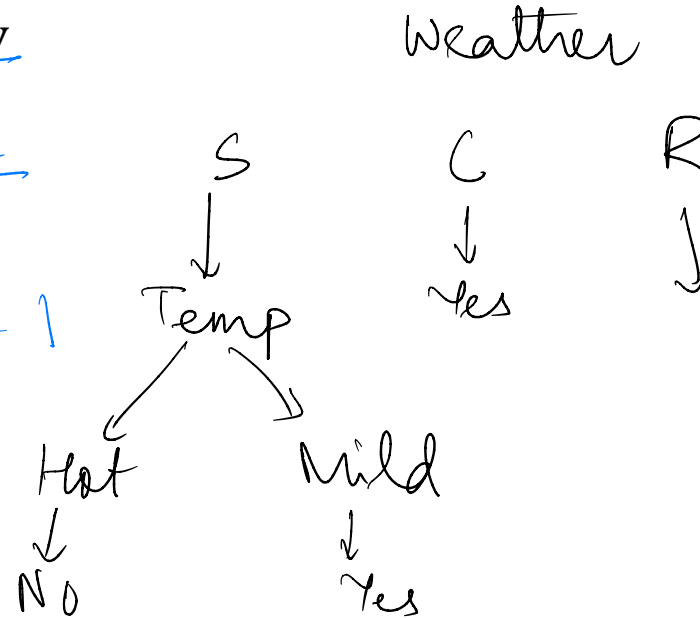
$p(\text{Yes}) = 0$ $p(\text{No}) = 1$

$G(\text{Hot}) = 0$ ←

Subset : Mild (1)

$p(\text{Yes}) = \frac{1}{1}$, $p(\text{No}) = \frac{0}{1}$

$G(\text{Mild}) = 0$ ←



Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 2: Internal nodes – Weather = Sunny

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 2: Internal nodes – Weather = Sunny

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 2: Internal nodes – Weather = Rainy

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 2: Internal nodes – Weather = Rainy

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



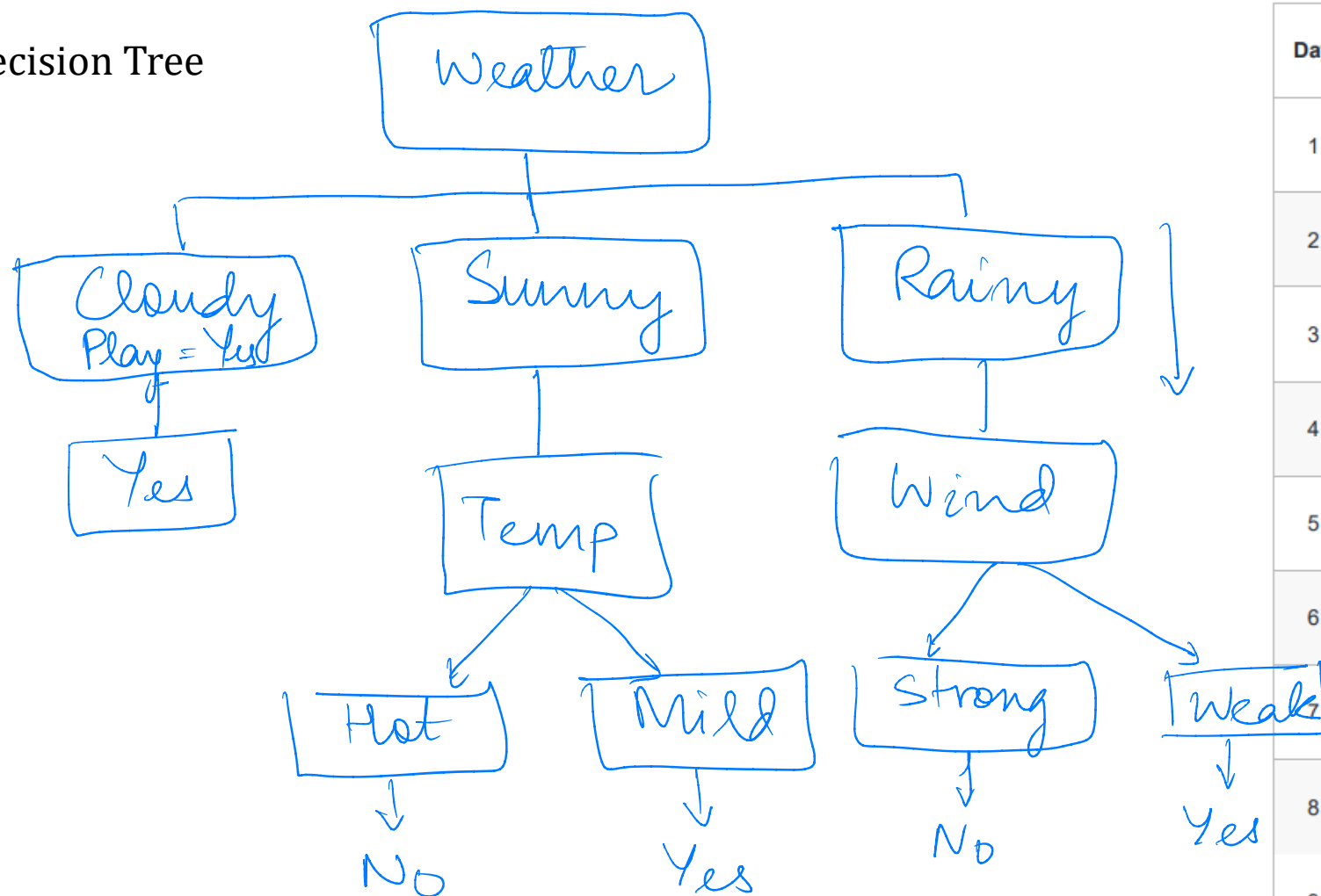
Step 2: Internal nodes – Weather = Rainy

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 3: Decision Tree

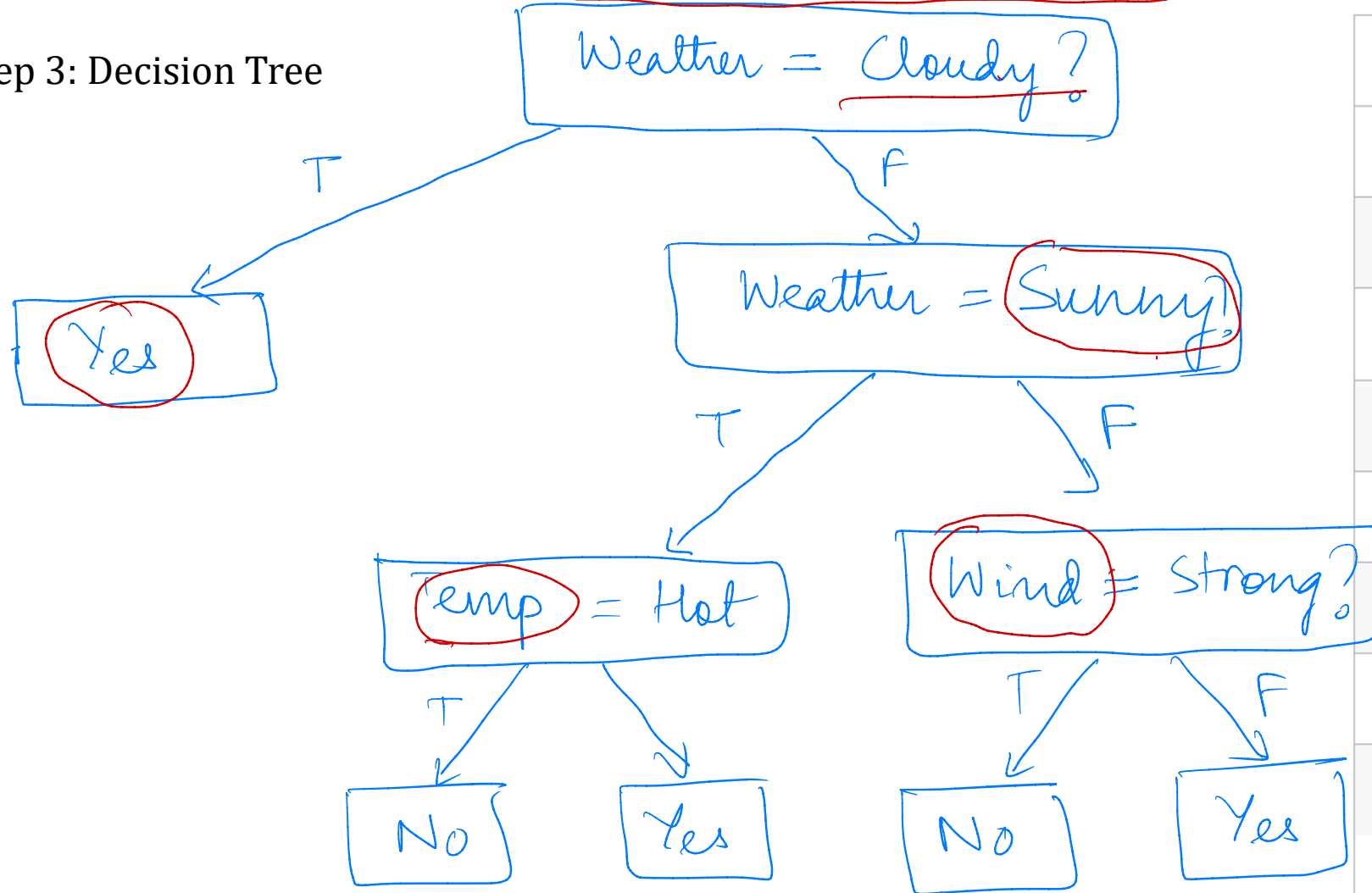


Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Decision Tree, Feature selection



Step 3: Decision Tree



Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

Revision of concepts

Revision of concepts



A decision tree is grown to full depth on a training set of 100 examples and achieves 100% training accuracy.

- (a) Why might such a fully grown tree perform poorly on unseen test data? What phenomenon does this illustrate?
- (b) Describe two strategies (other than pruning) to mitigate this issue when using decision trees in practice.
- (c) Explain how the Random Forest algorithm addresses the over-fitting problem in decision trees, mentioning both *bagging* and *feature randomization*.

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Thank you for your time!